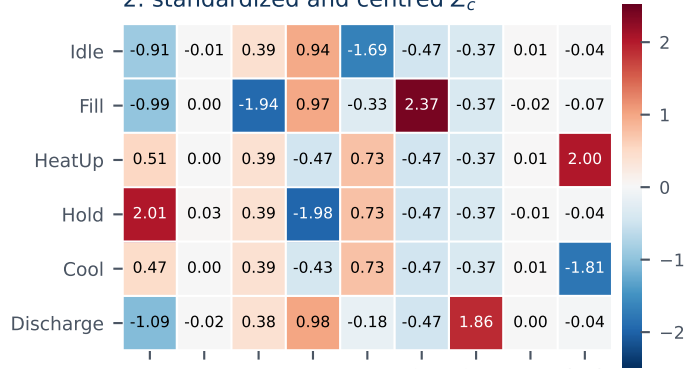
1: phase-wise means $M \in \mathbb{R}^{6 \times 9}$

	T	pH	κ	η	τ	Q_{in}	Q_{out}	P	dT/dt
Idle	10.4	6.65	4.80	2.19	0.20	0.00	0.00	1.20	0.00
Fill	8.87	6.65	3.90	2.20	0.65	1.50	0.00	1.20	-0.03
HeatUp	40.4	6.65	4.80	1.90	1.00	0.00	0.00	1.20	1.69
Hold	72.0	6.65	4.80	1.60	1.00	0.00	0.00	1.20	0.00
Cool	39.5	6.65	4.80	1.91	1.00	0.00	0.00	1.20	-1.46
Discharge	6.72	6.65	4.80	2.20	0.70	0.00	1.50	1.20	0.00

2: standardized and centred Z_c